PERSPECTIVE AND REFERENCE ARCHITECTURE

Adaptive Frames for Hybrid Intelligence

From Minsky's Knowledge Structures to Grounded, Probabilistic and Governed AI Agents

Central proposal. A computational frame should combine an ontology, validation constraints, uncertain expectations, active questions, executable procedures, transformations, provenance and repair operations. Neural models propose; semantic structures define; constraints verify; agents act; evidence and provenance determine whether the frame should be retained or revised.

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Abstract

Marvin Minsky's 1974 frame proposal argued that intelligence depends on retrieving and adapting large knowledge structures for stereotyped situations. Frames contain slots, constraints, defaults, procedural attachments, transformations and links to alternatives. The proposal anticipated contextual knowledge representation, script-like event models, active information seeking, explanation by expectation failure and the integration of declarative knowledge with control. It nevertheless remained an informal research programme: it did not supply a precise semantics, a scalable frame-selection algorithm, calibrated uncertainty, a learning procedure, data provenance or empirical evaluation.

This paper develops a contemporary successor: the *adaptive frame*, a goal-conditioned and temporally situated structure that joins RDF/OWL semantics, SHACL validation, probabilistic expectations, neural representations, retrieval-augmented generation, tool-using agents and provenance-aware repair. We define a formal frame tuple, derive a constrained frame-selection objective, describe a complete frame lifecycle and propose an evaluation protocol comparing language-model-only, ontology-only, retrieval-augmented and hybrid systems. A reproducible Google Colab simulation instantiates the architecture in a movie knowledge graph. It demonstrates competing frame hypotheses, SPARQL retrieval, SHACL validation, active questioning, human-confirmed repair, provenance capture and before/after quality visualisation. The resulting framework converts Minsky's enduring cognitive insight into an implementable and falsifiable research programme for trustworthy hybrid intelligence over the next decade.

Keywords: frames; knowledge graphs; semantic web; neuro-symbolic AI; SHACL; RDF; OWL; retrieval-augmented generation; agents; provenance; knowledge repair.

Take-home message

The paper's central claim is not that ontologies should replace neural models, or that language models should replace explicit knowledge. Intelligence requires a controlled cycle in which learned models propose contextual interpretations, semantic structures make their commitments explicit, constraints identify failure, tools obtain missing evidence and provenance-aware repair improves the system without silently rewriting history.

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1 Introduction

Artificial intelligence has repeatedly moved between two poles. Symbolic systems represent concepts and rules explicitly, permitting inspection and deduction, but often fail when the world is noisy, incomplete or unfamiliar. Statistical systems learn patterns from data and generalise across surface variation, but may produce answers that are unsupported, inconsistent or difficult to update. Foundation models intensify rather than remove this tension: their linguistic competence is broad, yet factual provenance, calibration, persistent identity and reliable constraint following remain open challenges [Bommasani et al., 2021, Ji et al., 2023].

Minsky's *A Framework for Representing Knowledge* was an early attempt to escape this binary [Minsky, 1975]. A frame is a remembered structure for a familiar situation. It contains terminals to be filled, conditions on possible values, defaults for incomplete observations, procedures for acquiring or checking values, and connections to alternative frames. A frame-system links multiple descriptions of the same entity or event, such as different visual viewpoints. Failure is not merely a negative outcome: a mismatch helps determine which alternative should be attempted next.

The proposal is often remembered as a precursor to object-oriented records or ontological classes. That interpretation is too narrow. A Minsky frame combines at least five functions that modern computing typically separates:

1. a **schema** describing entities and relations;
2. an **expectation model** containing typical or default values;
3. a **question generator** directing attention to missing evidence;
4. a **controller** specifying procedures that may be executed; and
5. a **repair mechanism** for explaining and responding to mismatch.

The central research question of this paper is therefore:

> *How can Minsky's contextual and procedural account of frames be converted into a formal, learnable, provenance-aware and empirically testable architecture using contemporary semantic data technologies and foundation models?*

The paper makes four contributions. First, it critically reconstructs the original frame proposal, separating its enduring insights from its unresolved assumptions. Second, it defines an adaptive-frame formalism that distinguishes hard constraints, soft expectations, defaults, provenance, tools and governance. Third, it presents a reference architecture grounded in RDF, OWL, SPARQL, SHACL, vector retrieval and agent actions. Fourth, it specifies a reproducible simulation and benchmark protocol through which the framework can be falsified, rather than remaining a conceptual metaphor.

2 Minsky's Frame Programme Reconstructed

2.1 Frames as large-grained knowledge structures

Minsky's initial complaint was representational granularity. Many early vision and problem-solving systems represented small facts or local features but lacked larger structures that could organise them. A frame supplies a global hypothesis: "this is a room", "this is a birthday party", or "this is a transaction". Observations are interpreted relative to the activated hypothesis, and the hypothesis specifies which observations matter.

This is not equivalent to forcing every observation into a rigid template. Frames can contain unfilled terminals, weak defaults and alternative branches. The frame reduces search by making some questions more relevant than others. Its value therefore lies as much in selective attention as in storage.

2.2 Terminals, markers and defaults

A terminal is a position that may be assigned a value. A room frame might contain terminals for walls, doors, windows, furniture and viewpoint. Terminal markers constrain what can fill a position. Some terminals refer to lower-level subframes, enabling hierarchical decomposition.

When observation does not provide a value, a default can be assigned. Defaults make efficient interpretation possible because most everyday situations are only partly observed. They also create epistemic risk. A typical value can be mistaken for a fact, and a historically common value can encode an unjustified social norm. The original proposal recognises that defaults may be displaced, but does not specify their confidence, provenance or legitimate scope.

2.3 Frame-systems and transformations

Related frames form systems. In the cube example, different views share terminals representing stable properties while viewpoint transformations alter the visible configuration. This anticipates multi-view representation: one entity may have visual, functional, causal, social and regulatory descriptions. No single description is sufficient for all goals.

The transformation idea is powerful but incomplete. A link labelled “move right” or “later” does not by itself explain the mechanism, temporal conditions or causal intervention. A modern version requires explicit event identity, time, persistence and action-conditioned transition models.

2.4 Language, discourse and scenarios

Minsky treats sentence interpretation as partial frame construction. Surface syntax activates semantic structures; successive sentences contribute to larger scene, thematic and narrative frames. Familiar scenarios create expectations about roles and event order. This is close to later script-based accounts [Schank and Abelson, 1977] and remains relevant to event extraction and narrative reasoning.

Crucially, an incomplete frame becomes a collection of questions. If an event frame expects an agent, object and destination, missing terminals specify what the system should seek next. Understanding is thus active: the representation controls information acquisition.

2.5 Memory, similarity and explanation

Memory retrieval is presented as problem solving, not as lookup. A cue activates candidate structures; partial matching preserves valuable assignments; mismatch may be explained by occlusion, damage, functional substitution or an exceptional context. Similarity networks connect frames through labelled differences, anticipating both knowledge-graph neighbourhoods and learned embedding spaces.

Minsky’s notion of an “excuse” is especially important. It is a structured explanation of why an expected feature is absent. Modern systems usually return a score or error label. A frame system should instead distinguish at least four possibilities: the observation is wrong, the selected frame is wrong, the case is a legitimate exception, or the representation is inadequate.

2.6 Control and procedural attachment

Frames are packets of data and processes. Activated frames establish local working environments in which procedures can inspect slots, obtain evidence and trigger alternatives. This avoids both extremes of a single central controller and completely independent local processes.

The idea prefigures tool-using agents, but it also creates a design hazard. If knowledge, inference and action are not cleanly separated, a corrupted assertion may trigger an unsafe operation. Contemporary implementations therefore need permission boundaries, audit trails and explicit distinctions between proposing, validating and executing an action.

3 Critical Assessment

3.1 What the paper got right

Table 1: Enduring contributions of the frame programme.

Contribution	Why it matters	Contemporary analogue
Contextual interpretation	Facts acquire relevance within goals and situations	Contextual retrieval and task-specific subgraphs
Defeasible defaults	Systems can operate under incomplete information	Priors, default rules and calibrated prediction
Multiple representations	One object supports several valid descriptions	Ontology modules, embeddings and multimodal views
Active questions	Missing slots direct information gathering	Active learning, agents and competency questions
Mismatch diagnosis	Failure supplies information about representation	SHACL reports, anomaly detection and model criticism
Procedural knowledge	Representation is connected to use and action	Tool calling, workflows and policy engines
Repair through experience	Learning modifies the representation, not only parameters	Knowledge-graph completion and ontology evolution

These insights challenge a persistent assumption: that intelligence can be reduced either to deduction over a fixed vocabulary or prediction from a fixed data distribution. Minsky instead treats representation as provisional, goal-sensitive and revisable.

3.2 Where the theory remains incomplete

3.2.1 Informal semantics

The original paper does not define when two frames are equivalent, when a slot assignment is licensed, or how conflicts between defaults should be resolved. Without model-theoretic or operational semantics, correctness depends on the implementer's intuition.

3.2.2 The selection circularity

A frame helps interpret observations after it has been selected, but selecting the correct frame appears to require an interpretation. Similarity and expectation can generate candidates, yet the theory lacks a principled ranking and rejection rule. Contemporary dense retrieval can narrow the candidate set, but semantic and empirical verification are still required.

3.2.3 Missing learning algorithm

Learning is described as debugging frames, but the theory does not determine when to add a slot, introduce an exception, split a frame, merge frames or forget outdated knowledge. These operations have different consequences and require evidence thresholds and governance.

3.2.4 Uncalibrated defaults

"Weak assignment" is not a quantitative uncertainty model. Probability, confidence, source reliability, constraint satisfaction and decision risk are distinct variables. Collapsing them into a single score recreates the problem Minsky identified: a number cannot preserve all the considerations that formed it.

3.2.5 Underdeveloped temporality and causality

Before-and-after frames encode succession but not necessarily causation. A modern account must represent events, timestamps, interventions, persistence, counterfactual alternatives and versioned beliefs. Otherwise, the system may mistake correlation or temporal order for a causal mechanism [Pearl, 2009].

3.2.6 No empirical test

The memo offers illustrations but no benchmark, baseline, complexity analysis, ablation or user study. Its explanatory reach is consequently greater than its demonstrated computational performance.

3.3 The critique of logic reconsidered

Minsky rightly objects to unrestricted classical deduction as a complete model of intelligence. Classical logic provides no native preference for relevant premises, treats conclusions monotonically and handles everyday exceptions poorly. Nevertheless, later description logics, default logics, truth-maintenance systems, probabilistic logics and paraconsistent approaches address important parts of this criticism [Brachman, 1977, Reiter, 1980, de Kleer, 1986].

The productive conclusion is not "logic failed". It is that logic should be assigned a bounded role. Formal semantics can define identities and necessary relationships; shapes can validate operational requirements; probabilistic models can rank plausible interpretations; and agents can obtain additional evidence.

4 From Frames to Semantic and Neuro-Symbolic Systems

4.1 RDF and OWL as the semantic substrate

RDF represents knowledge as subject–predicate–object triples [W3C RDF Working Group, 2014]. OWL adds formally interpreted classes, properties, individuals and restrictions [W3C OWL Working Group, 2012]. These technologies supply globally identifiable terms, interoperable graphs and machine-processable semantics. They provide part of a frame—principally its declarative schema— but not its full behaviour.

Under the open-world assumption, the absence of a triple does not entail that the corresponding statement is false. An OWL reasoner may therefore infer new class memberships without treating a missing release year as a data error. That distinction is essential: logical inference and operational completeness are different tasks.

4.2 SHACL as explicit verification

SHACL defines shapes against which RDF graphs can be validated [W3C RDF Data Shapes Working Group, 2017]. A shape can state that a movie requires exactly one title, that a duration is an integer within a range, or that a value belongs to an allowed set. A validation report identifies focus nodes, failed paths, severities and messages. This resembles Minsky's frame verification, but SHACL does not determine whether the data or the shape should be repaired.

4.3 Retrieval, embeddings and foundation models

Knowledge-graph embeddings and graph neural networks provide graded similarity over entities and relations [Nickel et al., 2016, Hamilton et al., 2017]. Retrieval-augmented generation combines parametric models with external memory, improving updateability and evidence access [Lewis et al., 2020]. Graph-based retrieval can additionally organise entity communities and global themes [Edge et al., 2024].

These methods help solve candidate generation but do not by themselves establish semantic correctness. Similar vectors need not denote equivalent entities, and a retrieved passage can still be false, outdated or inapplicable to the current context.

4.4 Neuro-symbolic learning and agents

Neuro-symbolic systems combine learned representations with logical or relational knowledge. Logic Tensor Networks, for example, ground logical expressions into differentiable computations [Badreddine et al., 2022]. Such systems offer a mechanism for soft constraint satisfaction, but a complete adaptive frame also requires provenance, context, temporal change and explicit repair.

Tool-using language-model agents interleave reasoning with environment actions [Yao et al., 2023]. This operationalises attached procedures, yet unrestricted agents can propagate hallucinations into external systems. A frame should therefore mediate tool use: candidate actions are proposed, checked against semantic and governance constraints, executed with limited authority, and then recorded with their effects.

5 The Adaptive-Frame Formalism

5.1 Definition

An adaptive frame at time t is defined as

$$\mathcal{F}_t = \langle I, S, C^H, C^S, D, Q, P, T, \Pi, \Gamma \rangle, \tag{1}$$

where:

- I identifies the frame, its semantic type and applicable context;
- S is a set of slots, relations and current assignments;
- C^H contains hard logical, safety or permission constraints;
- C^S contains weighted, defeasible or probabilistic constraints;
- D contains defaults and learned priors;
- Q contains questions capable of obtaining missing evidence;
- P contains permitted procedures and tool policies;
- T contains temporal, spatial and causal transformations;
- Π records evidence, confidence, source and derivation; and
- Γ specifies governance, access, risk and escalation rules.

The definition intentionally keeps validity, likelihood and authority separate. A statement can be syntactically valid but unlikely, likely but unsupported, or well supported but unavailable to a particular user.

5.2 Frame selection

Let $\mathcal{O}_t$ denote current observations and $\mathcal{G}_t$ the active goal. Candidate frames are retrieved from a library $\mathbb{F}$ using symbolic keys, graph neighbourhoods and vector similarity. Selection is then defined as

$$\begin{aligned}\mathcal{F}_t^* = \arg\max_{\mathcal{F}\in\mathbb{F}} \Big[& \alpha \log p(\mathcal{O}_t \mid \mathcal{F}, \mathcal{G}_t) + \beta\, \mathrm{Fit}_{\mathrm{sem}}(\mathcal{F}, \mathcal{O}_t) + \gamma U(\mathcal{F}, \mathcal{G}_t) \\ & - \delta\, \mathrm{Violation}(\mathcal{F}, \mathcal{O}_t) - \eta\, \mathrm{Risk}(\mathcal{F}) - \kappa\, \mathrm{Cost}(\mathcal{F}) \Big],\end{aligned} \tag{2}$$

subject to

$$C^H(\mathcal{F}, \mathcal{O}_t) = \mathrm{true}. \tag{3}$$

The terms serve distinct purposes:

- $p(\mathcal{O}_t \mid \mathcal{F}, \mathcal{G}_t)$ measures evidential likelihood;
- $\mathrm{Fit}_{\mathrm{sem}}$ measures type and relation compatibility;
- U measures expected usefulness for the current goal;
- Violation measures failed shapes or soft rules;
- Risk penalises harmful interpretations or actions; and
- Cost accounts for computation, tools and human effort.

For a teaching implementation, each term can be normalised to $[0, 1]$ and a weighted sum used. A research implementation should learn or calibrate the weights and report sensitivity analyses.

5.3 Soft satisfaction

If $C^S = \{c_1, \ldots, c_m\}$ and each constraint has weight w_i and satisfaction $s_i \in [0, 1]$, then

$$\mathrm{SoftSat}(\mathcal{F}) = \frac{\sum_{i=1}^{m} w_i s_i(\mathcal{F})}{\sum_{i=1}^{m} w_i}. \tag{4}$$

Unlike a hard SHACL violation, a low soft score should reduce confidence without automatically rejecting an unusual but possible observation.

5.4 Active question selection

An unfilled slot generates a set of candidate questions $\mathcal{Q}$. Let $H(\mathcal{F})$ be uncertainty over the competing frames. The next question is

$$q^* = \arg\max_{q \in \mathcal{Q}} \frac{\mathbb{E}\left[H(\mathcal{F}) - H(\mathcal{F} \mid A_q)\right] \times \mathrm{Relevance}(q, \mathcal{G}_t)}{\mathrm{Cost}(q) + \epsilon}, \tag{5}$$

where A_q is the possible answer. High-value questions discriminate between frames or resolve constraints; low-value questions merely collect detail.

5.5 Repair

Let V_t be a structured validation and contradiction report, and H_t the repair history. A repair operator produces

$$\mathcal{F}_{t+1} = \mathcal{R}(\mathcal{F}_t, \mathcal{O}_t, V_t, H_t). \tag{6}$$

Permitted operations include filling a slot, correcting a literal, mapping an identifier, specialising or generalising a type, changing frame, adding a scoped exception, querying a source, requesting human confirmation or abstaining. The system should never silently replace the original assertion. The replacement, source, responsible agent and justification form a provenance chain.

6 Reference Architecture

6.1 Layer 1: observations and grounding

The observation layer ingests text, tables, existing RDF, images or sensor measurements. Extractors generate candidate entities and relations while preserving links to source spans or records. Multimodal encoders may ground visual and linguistic patterns in a shared space, but symbolic identifiers are created only when identity criteria are satisfied.

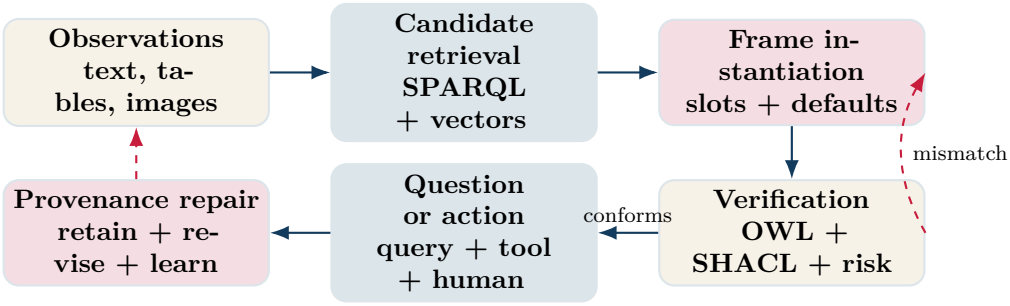


Figure 1: Adaptive-frame lifecycle. The architecture is cyclic: validation failure prompts evidence gathering or representation repair rather than a forced answer.

6.2 Layer 2: candidate retrieval

Candidate retrieval uses complementary mechanisms:

- exact identifiers and ontology types;
- SPARQL graph patterns;
- graph neighbourhoods and community summaries;
- dense vector similarity; and
- episodic memory of previous repairs.

Vector retrieval generates hypotheses; it does not establish equivalence. Symbolic retrieval supplies precision; it can fail when vocabulary differs. Hybrid retrieval is expected to provide better recall without abandoning semantic verification.

6.3 Layer 3: semantic frame

The RDF/OWL graph supplies persistent identifiers, type hierarchies and relations. Named graphs or contextualised statements distinguish sources, versions and scenarios. Ontology modules allow several compatible perspectives instead of one monolithic classification.

6.4 Layer 4: verification

Verification combines four tests:

1. **logical consistency**: whether asserted axioms can be jointly interpreted;
2. **shape conformance**: whether operational data requirements are met;
3. **probabilistic plausibility**: whether the observation is expected;
4. **governance**: whether the proposed use is authorised and sufficiently safe.

The outputs must remain separate. A rare observation may be perfectly valid; an internally consistent graph may contain false assertions; and a true fact may still be unavailable for a particular purpose.

6.5 Layer 5: questions and actions

If required evidence is missing, the frame selects a question or tool. In a low-risk teaching system this may be a SPARQL query. In a clinical or physical system, tool access must be constrained by role, purpose and uncertainty. Interleaving reasoning and actions improves access to external information [Yao et al., 2023], but the adaptive-frame controller adds pre-action validation and post-action provenance.

6.6 Layer 6: repair memory

Every proposed repair is represented as an activity linking the original and revised assertions. PROV-O concepts such as Entity, Activity, Agent, `wasDerivedFrom` and `wasRevisionOf` support this record [W3C Provenance Working Group, 2013]. Recurrent repairs become candidates for ontology or shape revision, but widespread repetition alone does not justify a semantic change.

7 Mapping to the Semantic Data Technologies Pipeline

Table 2: Operational mapping from Minsky's vocabulary to the SDT toolchain.

Frame concept	**Implementation**	**Important qualification**
Frame	OWL class + SHACL node shape + policy object	An OWL class alone is not executable
Terminal	RDF property or SHACL property path	Missing does not imply false under open-world semantics
Marker	Range/restriction/shape constraint	Inference and validation answer different questions
Default	Scoped rule or probabilistic prior	Do not assert defaults as observed facts
Question	Competency question or SPARQL template	Rank by relevance and information gain
Alternative	Competing type/frame hypothesis	Preserve scores and rejection reasons
Transformation	Event or state-transition resource	Attach time, action and provenance
Matching	SPARQL + embedding retrieval	Similarity is not identity
Excuse	Diagnosed validation explanation	Explanation may implicate data or model
Procedure	Python function or agent tool	Apply permissions and audit logging
Repair	Versioned RDF update with PROV-O	Preserve the original claim

The executable pipeline follows twelve student-facing steps:

1. install and import Python packages;
2. load or construct a movie ontology;
3. inspect classes, properties and individuals;
4. visualise the graph;
5. define candidate frames;
6. score competing frame hypotheses;
7. express competency questions in SPARQL;
8. define SHACL shapes;
9. validate and parse the report;
10. select the most informative missing question;
11. apply a transparent, human-confirmed repair; and
12. record provenance and export all artefacts.

8 Worked Movie-Knowledge-Graph Simulation

8.1 Scenario

Consider an imported record for *The Walking Dead*. The source labels it as a movie, supplies a title and genres, but omits a release year and includes episode-related evidence. Three candidate frames are considered: `MovieFrame`, `SeriesFrame` and `DocumentaryFrame`.

Each frame specifies required slots and characteristic evidence. The series frame receives support from properties such as season count or episode count; the movie frame receives support from a running-time value; documentary receives support from a documentary genre or factual subject relation.

8.2 Transparent score

For a simple teaching model, define

$$\text{score}(F) = 0.45\ \text{type}(F) + 0.35\ \text{slot}(F) + 0.20\ \text{cue}(F) - 0.25\ \text{violation}(F). \tag{7}$$

All components are reported separately. The visible decomposition matters more than optimisation: students can see why an initially asserted type may lose to an alternative frame.

8.3 SHACL validation

A simplified movie shape can be written as:

Listing 1: Python construction of two SHACL requirements.

```
movie_shape.add((EX.MovieShape, SH.targetClass, EX.Movie))

# Every Movie should have one human-readable title.
title_rule = BNode()
movie_shape.add((EX.MovieShape, SH.property, title_rule))
movie_shape.add((title_rule, SH.path, EX.title))
movie_shape.add((title_rule, SH.minCount, Literal(1)))

# In this operational dataset, a release year is required.
year_rule = BNode()
movie_shape.add((EX.MovieShape, SH.property, year_rule))
movie_shape.add((year_rule, SH.path, EX.releaseYear))
movie_shape.add((year_rule, SH.minCount, Literal(1)))
movie_shape.add((year_rule, SH.datatype, XSD.gYear))
```

A missing year produces a validation result. The result does not prove that the resource is not a movie. It states that the data fail the chosen operational shape. The controller can respond by asking for a year, querying an approved source, weakening the shape with justification, or choosing an alternative frame.

8.4 Active question and repair

Suppose the two highest-scoring frames are Movie and Series. The most useful question is not necessarily "Who acted in it?" because both frames may contain actors. "How many seasons does it have?" has greater discriminative value. After a verified season count is supplied, the system proposes a type revision. A human confirms the change, and the graph records:

- the original classification;
- the revised classification;
- the evidence used;
- the person or software agent approving the revision;

- the time of revision; and
- the validation results before and after repair.

Critical point

Automatic repair should be conservative. Literal normalisation—for example, converting an unambiguous numeric string into an integer—may be safe. Semantic reclassification from Movie to Series should normally be proposed, explained and confirmed rather than silently executed.

9 Evaluation Framework

9.1 Research questions

RQ1 Does hybrid frame selection improve semantic correctness under incomplete and noisy observations?

RQ2 Does active questioning reduce the number of observations required to select an appropriate frame?

RQ3 Does provenance-aware repair improve adaptation without increasing unsupported assertions?

RQ4 Which components contribute most under distribution and schema shift?

RQ5 Do structured mismatch explanations improve human error detection and appropriate trust?

9.2 Baselines

At least four systems should be compared:

1. a language-model-only classifier or answer generator;
2. an ontology and rule system without learned retrieval;
3. vector RAG or GraphRAG without explicit frame repair; and
4. the complete adaptive-frame architecture.

An oracle system with complete observations provides an upper reference but should not be confused with a deployable baseline.

9.3 Datasets and perturbations

The benchmark should contain static and evolving domains. The movie graph is a transparent teaching case. A stronger evaluation should add an operational domain and a high-stakes case such as clinical-study eligibility.

Controlled perturbations should include:

- missing required properties;
- misspelled labels and duplicate identifiers;
- incorrect datatypes and units;
- contradictory source assertions;
- temporal updates;
- novel subtypes;
- misleading retrieved documents; and
- changes to schema and governance rules.

9.4 Metrics

Table 3: Proposed evaluation dimensions and metrics.

Dimension	Metric	Interpretation
Candidate retrieval	Recall@k, MRR	Is the appropriate frame retrieved?
Frame selection	Accuracy, macro-F1	Is the most suitable frame selected?
Slot completion	Precision, recall, F1	Are missing values completed correctly?
Calibration	Brier score, ECE	Does confidence match observed correctness?
Validation	Violations detected per injected fault	Are operational defects found?
Repair	Correct and harmful repair rates	Does revision improve the graph safely?
Provenance	Supported-claim coverage	Can conclusions be traced to evidence?
Adaptation	Performance after shift	Does the system survive new contexts?
Action	Success and unsafe-action rate	Are tools used effectively and safely?
Human factors	Error detection, time, trust calibration	Do explanations help users?
Efficiency	Latency, token/tool cost	Is the improvement operationally viable?

9.5 Ablation studies

Remove one component at a time: ontology, SHACL, vector retrieval, defaults, active questions, provenance, repair memory, temporal context or governance. An ablation is informative only when evaluated under the condition that the removed component addresses. For example, provenance should be tested under source conflict, not only clean classification.

9.6 Statistical analysis

For binary task success, a mixed-effects logistic model can account for repeated instances and domains:

$$\operatorname{logit} \Pr(Y_{ij} = 1) = \beta_0 + \beta_1 \,\mathrm{Hybrid}_i + \beta_2 \,\mathrm{Shift}_j + \beta_3 (\mathrm{Hybrid}_i \times \mathrm{Shift}_j) + u_{\mathrm{domain}}. \tag{8}$$

For paired systems, bootstrap confidence intervals should be reported for differences in F1, calibration and violation rate. Multiple seeds, model versions and prompt templates should be retained as experimental factors rather than discarded as nuisance variation.

10 Clinical Eligibility as a High-Stakes Extension

Consider a patient being evaluated for a clinical study. The frame contains slots for age, diagnosis, biomarker status, previous therapy, renal function, pregnancy status and consent. Some criteria are hard exclusions; others depend on context or clinician judgement.

A language model can extract candidate facts from a note, but should not make the eligibility decision alone. The adaptive frame performs the following:

1. map note phrases to ontology concepts;
2. preserve links to the source sentences and laboratory records;
3. distinguish observed values from defaults and predictions;
4. validate required evidence and units;
5. calculate probabilistic uncertainty where appropriate;
6. ask for missing discriminative information;
7. apply protocol-specific rules;
8. provide an explanation and abstain when unresolved; and
9. require authorised human confirmation.

This example reveals why a single score is insufficient. A patient may have a high predicted probability of eligibility while lacking a mandatory test. The appropriate output is "potentially eligible, evidence incomplete", not an unqualified yes.

11 Governance, Ethics and Security

11.1 Defaults as institutional claims

Defaults are not neutral conveniences. They encode what designers consider normal, typical or safe. Every default should therefore carry:

- its applicable population and context;
- its empirical or policy source;
- its creation and review dates;
- confidence and known exceptions;
- an accountable owner; and
- a mechanism for challenge and revision.

11.2 Ontology and frame poisoning

Attackers may alter definitions, introduce misleading equivalences, manipulate retrieval sources or create repair histories that normalise false claims. Signed artefacts, source allowlists, review workflows, version comparison and anomaly detection are therefore required. Tool permissions should follow least privilege.

11.3 Explanations and appropriate trust

An explanation should distinguish evidence, inference, default and policy. A plausible natural-language rationale is not sufficient. Faithful explanations must be generated from the actual frame-selection, validation and repair trace. Human-centred evaluation should measure whether explanations help users detect errors, not merely whether users find them satisfying [Ribeiro et al., 2016, Amershi et al., 2019].

12 Research Agenda for the Next Five to Ten Years

12.1 Learnable but stable frames

Future systems must learn new frames while protecting stable identifiers and critical constraints. Promising directions include modular ontology evolution, continual learning with semantic regularisation, and repair proposals reviewed through human-in-the-loop workflows.

12.2 Multimodal grounding and world models

Predictive world models can learn temporally structured representations from video and action. Adaptive frames can supply explicit entity identity, role, constraint and provenance around those latent representations. The open problem is bidirectional alignment: symbols must be grounded in perception, while latent changes must be expressible as inspectable events.

12.3 Causal and counterfactual frames

Transformations should evolve from labelled links into causal mechanisms. Future frames should represent interventions, competing mechanisms and counterfactual outcomes. Causal assumptions must remain explicit because no amount of observational scale removes the need to state them.

12.4 Collective and contested frames

Different communities may use incompatible but locally coherent classifications. A robust system should not erase disagreement by merging all assertions into a single graph. It should represent viewpoints, authority, scope and negotiated alignments. Multi-agent systems then require protocols for proposing, challenging and revising shared frames.

12.5 Frames as scientific instruments

Adaptive frames can become executable scientific hypotheses. A frame predicts which entities, relationships and observations should occur. A validation failure becomes a model criticism; repeated, well-supported failures motivate theory revision. This connects semantic data technologies to reproducible science rather than treating ontologies only as documentation.

13 Discussion

The adaptive-frame proposal deliberately avoids claiming that a single formalism can represent all intelligence. RDF and OWL are suited to persistent identity and shared semantics; SHACL to operational validation; probabilistic models to uncertainty; embeddings to graded retrieval; foundation models to language and pattern induction; and agents to evidence gathering and action. The frame is the coordinating unit, not a replacement for these components.

This separation also clarifies several frequent confusions. A missing triple is not false. A SHACL violation is not necessarily a factual error. A probable value is not an observed value. A similar embedding is not the same entity. A logically consistent assertion is not necessarily true. An explanation is not faithful merely because it sounds plausible. A repair is not legitimate unless its source, scope and authority are recorded.

The proposal has limitations. The formal objective contains terms that must be operationalised and calibrated. Large frame libraries may make candidate retrieval costly. Human confirmation can become a bottleneck. Ontologies may be wrong, and formal constraints can institutionalise harmful assumptions. Some domains may be better served by lightweight schemas than rich ontologies. Finally, the current Colab simulation is pedagogical; it demonstrates the architecture but cannot establish superiority over production systems.

Nevertheless, the framework is falsifiable. If hybrid frame systems do not improve constraint satisfaction, adaptation, provenance or appropriate human trust relative to simpler baselines, their additional complexity is not justified.

14 Conclusion

Minsky's frame paper remains valuable because it treats representation as an active, contextual and revisable component of intelligence. Its enduring lesson is that knowledge is more than a collection of propositions: it includes expectations, questions, transformations, procedures and responses to failure.

The original theory remained informal and was never adequately evaluated. This paper converts it into an adaptive-frame architecture integrating semantic graphs, constraints, probabilistic expectations, neural retrieval, tool use, provenance and repair. Neural models propose; ontologies define; shapes validate; probabilities rank; agents obtain evidence; governance constrains; and repair memory enables learning without silently rewriting the past.

The most important future benchmark is therefore not whether a system can answer a familiar question. It is whether the system can recognise when its current frame is inadequate, explain the mismatch, acquire the right evidence and revise its representation safely.

Data and code availability

The accompanying Google Colab notebook constructs a self-contained movie knowledge graph and may optionally load an externally supplied OWL file. It exports Turtle, RDF/XML, SHACL reports, provenance records, CSV tables and PNG figures. The notebook and manuscript source are distributed with the ARU–CJLU Semantic Data Technologies course repository.

Conflict of interest

The author declares no conflict of interest.

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A Algorithmic Summary

Algorithm 1: Adaptive-frame interpretation and repair

1. **Input:** observations $\mathcal{O}_t$, goal $\mathcal{G}_t$ and frame library $\mathbb{F}$.
2. Retrieve candidate frames K using symbolic and vector indexes.
3. For each $F \in K$, instantiate observed slots and explicitly label defaults.
4. Compute semantic fit, soft satisfaction, cost and risk; validate hard constraints.
5. Rank candidates using equation (2) and select F^*.
6. If F^* satisfies the action threshold, execute only the least-privileged permitted action and record its effects.
7. Otherwise select q^* using equation (5); obtain trusted evidence, ask a human, or abstain.
8. If a repair is approved, apply $\mathcal{R}$ while preserving the original graph and provenance.
9. Revalidate and report the before/after difference.

B Competency Questions for the Movie Demonstration

1. Which resources are currently typed as movies?
2. Which required properties are missing for each movie?
3. Which resources contain evidence characteristic of television series?
4. Which asserted types conflict with the highest-scoring candidate frame?
5. Which values were inferred, defaulted, observed or manually confirmed?
6. Which validation issues were resolved by each repair?
7. Who or what approved a semantic reclassification?
8. Can the previous graph state be reconstructed?

C Reproducibility Checklist

- Run the notebook from top to bottom in a fresh Colab runtime.
- Record package versions and random seeds.
- Keep the original graph separate from the repaired graph.
- Export the shapes graph and both validation reports.
- Preserve frame-score components, not only the winning label.
- Distinguish human-confirmed repairs from automated normalisation.
- Report failed and abstained cases.
- Evaluate under controlled missingness, contradiction and schema shift.